

GOSFORD HIGH SCHOOL



SOLNS

Year 12 HSC Mathematics Extension 1

Assessment Task #3

Time Allowed: 70 Minutes

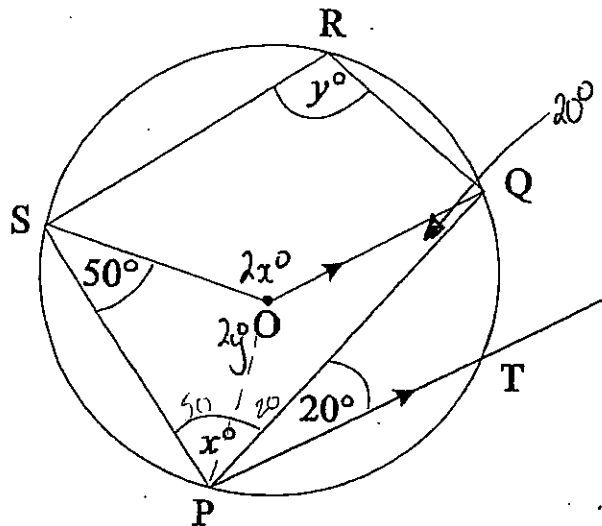
Instructions:

- Attempt all questions.
- Start each question on a new sheet of paper.
- All questions are of equal value.
- Board approved calculators may be used.
- Write using black or blue pen.
- All necessary working should be shown in every question.
- A table of standard integrals is provided at the back of this paper.

Question 1 (12 marks) (Start a new sheet of paper.)

Marks

- a. On the diagram, O is the centre of the circle and $PT \parallel OQ$. P, Q, R and S are points on the circle. Find the values of x and y , giving reasons.

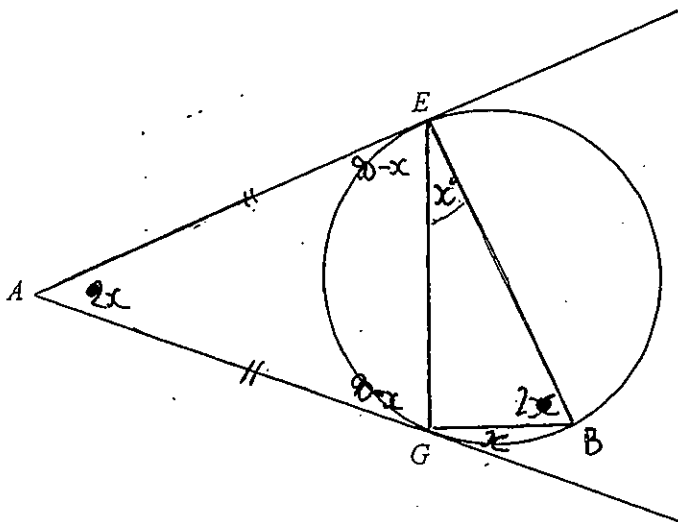


$$\begin{aligned} x + 2y &= 290 \quad - 1 \\ x + y &= 180 \quad - 2 \\ \hline y &= 110 \\ x &= 70 \end{aligned}$$

4

Diagram not to scale

- b. AE and AG are tangents to a circle. B is a point on the circle such that $\angle EBG$ and $\angle EAG$ are equal and are both double $\angle GEB$. Let $\angle GEB = x^\circ$.



$$2x = 90 - x \quad (\text{angles})$$

$$x = 30$$

$$\therefore \angle EGB = 60^\circ$$

4

Diagram not to scale

- Find x , giving reasons for your answer.
- Hence, prove that EB is a diameter of the circle.

- c. In the diagram below, AC is the diameter of circle $AECF$ with centre O and BD is a tangent to the circle at C .

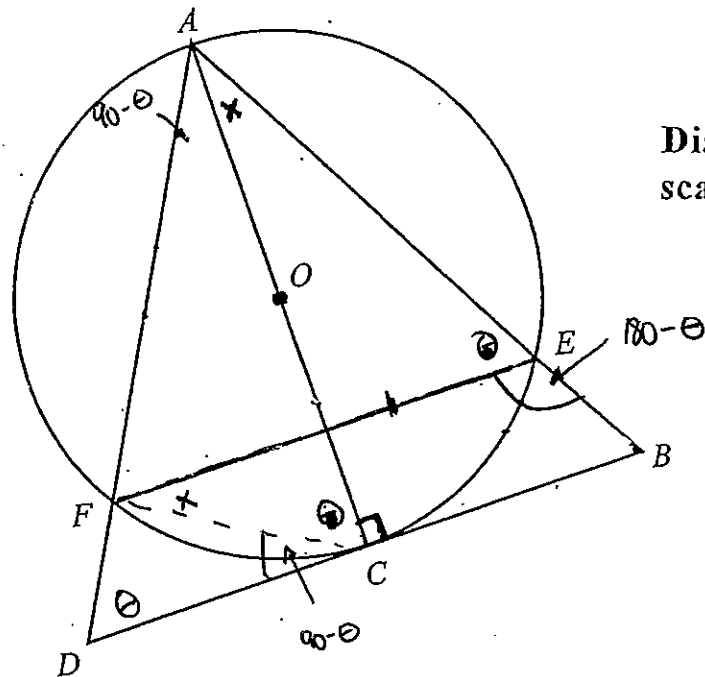


Diagram not to scale

4

- Neatly copy the diagram on to your answer sheet.
- Prove that $DFEB$ is a cyclic quadrilateral.

END OF QUESTION 1

Question 2 (12 marks) (Start a new sheet of paper.)

	Marks
a. Ryan started a new job for which his starting salary was \$52 000 pa for the first year. As an incentive he receives an increment of \$2500 following each year of service for the first ten years and \$3500 for each year of service after that.	
i. Find the amount of salary Ryan will earn in his 18 th year of service.	1
ii. What is Ryan's total earnings for the first 18 years of service?	3
b. A farmer borrows \$80 000 to purchase new machinery. The interest is calculated monthly at the rate of 2% per month and is compounded monthly.	
The farmer intends to repay the loan over two years making payments of \$M each month.	
i. Show that at the end of the first month the farmer owes (in \$)	1
$80\,000(1.02) - M$	
ii. Deduce that the amount owing at the end of two years is given by (in \$)	2
$80\,000(1.02)^{24} - M(1 + 1.02^2 + 1.02^3 + \dots + 1.02^{23})$	
iii. Find the amount of each monthly repayment.	2
c. Abbie was born on 19 th December 2006. On that day her grandparents opened a trust account by depositing \$2000 immediately at birth and \$250 each year thereafter on her birthday.	
Abbie is to receive the accumulated value from this account as a 21 st Birthday present on the 19 th December 2027 immediately after a final payment has been made.	
The interest earned on the account is 8% pa compounded every six months.	
i. Show that the initial deposit accumulated to \$10 385.57 after 21 years.	1
ii. How much did Abbie's grandparents give her on her 21 st Birthday?	2

END OF QUESTION 2

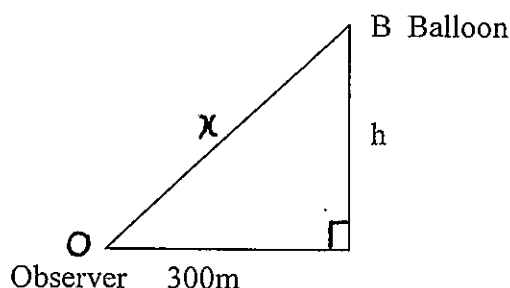
Question 3 (12 marks) (Start a new sheet of paper.)

Marks

- a. A bath tub which holds 240 litres, when full, is drained so that at time t seconds the volume of water V , in litres, is given by:

$$V = 240\left(1 - \frac{t}{60}\right)^2 \text{ for } 0 \leq t \leq 60$$

- i. After how many seconds was the bathtub one-quarter full? 2
- ii. At what rate was the water draining out when the bathtub was one-quarter full? 2
- b. A weather balloon is rising vertically at 5m/s. An observer is standing on the ground 300m from the point where the balloon was released, as indicated in the diagram. Let h metres be the vertical height of the balloon above the ground and x metres be the distance between the observer and the balloon at any time t secs.



- i. Show that $x = \sqrt{h^2 + 90\,000}$ 1
- ii. At what rate is the distance between the observer and the balloon changing when the balloon is 400m high? 2
- c. N is the number of kangaroos in a certain population at time t years.
- The population size N satisfies the equation
- $$\frac{dN}{dt} = -k(N - 500) \text{ (where } k \text{ is a constant)}$$
- i. Verify that $N = 500 + Ae^{-kt}$ (where A is a constant) is a solution of the equation. 1
- ii. If initially there are 3500 kangaroos and after 3 years there are 3300 find the values of A and k . 2
- iii. Find, to the nearest year, when the number of kangaroos begins to fall below 2300. 2

END OF QUESTION 3

Question 4 (12 marks) (Start a new sheet of paper.)

Marks

- a. The velocity V m/s of a particle moving in a straight line is given by

$$V = \sqrt{4x^5 - 20x^3 + 40x}$$

- i. Find an expression for acceleration in terms of x .

1

- ii. Hence, find the maximum speed of the particle.

2

- b. A particle moves in a straight line from a fixed point, O , along the x axis. Its velocity V m/s can be found using:

$$V^2 = x^2 - 9$$

Initially the particle is located 5m to the right of O and is moving towards the origin.

- i. Find the acceleration of the particle in terms of x .

1

- ii. Show that the particle does not pass through the origin.

1

- iii. Find the set of possible values of x .

2

- iv. Describe the motion of the particle.

1

- c. The acceleration of a particle moving in a straight line is given by:

$$a = -16e^{-4x}$$

Initially the particle is at the origin with velocity $2\sqrt{2}$ m/s.

- i. Prove that $v = 2\sqrt{2}e^{-2x}$

2

- ii. Hence, show that $x = \frac{1}{2} \ln(1 + 4\sqrt{2}t)$.

2

END OF ASSESSMENT TASK

STANDARD INTEGRALS

$$\int x^n dx = \frac{1}{n+1} x^{n+1}, \quad n \neq -1; \quad x \neq 0, \text{ if } n < 0$$

$$\int \frac{1}{x} dx = \ln x, \quad x > 0$$

$$\int e^{ax} dx = \frac{1}{a} e^{ax}, \quad a \neq 0$$

$$\int \cos ax dx = \frac{1}{a} \sin ax, \quad a \neq 0$$

$$\int \sin ax dx = -\frac{1}{a} \cos ax, \quad a \neq 0$$

$$\int \sec^2 ax dx = \frac{1}{a} \tan ax, \quad a \neq 0$$

$$\int \sec ax \tan ax dx = \frac{1}{a} \sec ax, \quad a \neq 0$$

$$\int \frac{1}{a^2 + x^2} dx = \frac{1}{a} \tan^{-1} \frac{x}{a}, \quad a \neq 0$$

$$\int \frac{1}{\sqrt{a^2 - x^2}} dx = \sin^{-1} \frac{x}{a}, \quad a > 0, \quad -a < x < a$$

$$\int \frac{1}{\sqrt{x^2 - a^2}} dx = \ln \left(x + \sqrt{x^2 - a^2} \right), \quad x > a > 0$$

$$\int \frac{1}{\sqrt{x^2 + a^2}} dx = \ln \left(x + \sqrt{x^2 + a^2} \right)$$

NOTE: $\ln x = \log_e x, \quad x > 0$

GHS - MATHEMATICS EXTENSION 1

ASSESSMENT TASK #3 SOLUTIONS

QUESTION 1.

a. $\angle QP = 20^\circ$ (alternate $\angle$'s in $\parallel$ arcs)
 reflex $\angle SOQ = 2x^\circ$ ($\angle$ at the centre of O is twice $\angle$ at circumference standing on same arc)

$\therefore \angle SOQ = 360^\circ - 2x^\circ$ ($\angle$'s at revolution = 360°)

$\therefore 50^\circ + x^\circ + 20^\circ + (360^\circ - 2x^\circ) = 360^\circ$
 ($\angle$ sum of quadrilateral = 360°)

$$\therefore 430 - x = 360$$

$$70 = x$$

$y = 110$ (opposite $\angle$'s in cyclic quadrilateral are supplementary.)

b. (i) $\angle EBG = \angle EAG = 2x^\circ$ (given)

$\therefore$ Since $\triangle AEG$ is isosceles.

(tangents drawn from an external point are equal)

$\angle AEG = 90^\circ - x^\circ$ (equal base $\angle$'s in isosceles $\triangle$, $\angle$ sum of $\triangle = 180^\circ$)

$\therefore \angle EBG = 90^\circ - x^\circ$ ($\angle$ between chord and tangent is $\angle$ in the alternate segment)

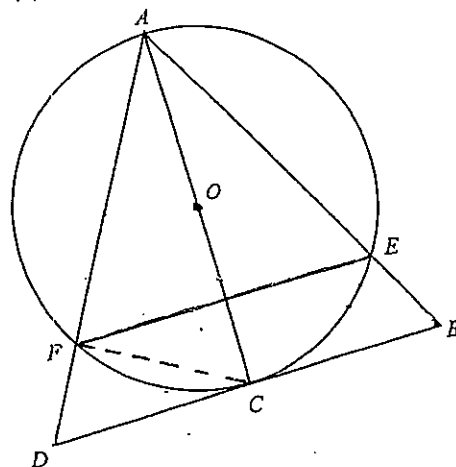
$$\therefore 2x = 90 - x$$

$$3x = 90$$

$$x = 30$$

(ii) $\angle EGB = 90^\circ$ ($\angle$ sum of $\triangle = 180^\circ$)

c. (i)



(ii) Construct a line interval FC.

Let $\angle ACF = x^\circ$

$\therefore \angle AEF = x^\circ$ ($\angle$'s at the circumference standing on the same arc are equal)

$\therefore \angle BEF = 180^\circ - x^\circ$ (straight $\angle$)

Now $\angle AFC = 90^\circ$ ($\angle$ in semicircle = 90°)
 (AC diameter given)

$\therefore \angle CFD = 90^\circ$ (straight $\angle$)

and $\angle FCD = 90^\circ - x^\circ$ ($\angle$ between tangent and a radius (AC is diameter) = 90°)

$\therefore \angle FDC = x^\circ$ ($\angle$ sum of $\triangle FCD = 180^\circ$)

Since $\angle FDC$ and $\angle BEF$ are supplementary DFEB is a cyclic quadrilateral (opposite $\angle$'s in cyclic quadrilateral are supplementary)

QUESTION 2

a.

(i) note that in first ten years there are 9 increments

$$\therefore T_{18} = 52000 + 9 \times 2500 + 8 \times 3500 = 102500$$

$\therefore$ In his 18th year Ryan will receive a salary of \$102 500

(ii) Ryan's total earnings $\Rightarrow A_1 + A_2$

$$(T_1 - T_0)A_1 = 52000 + 52000 + 1 \times 2500 + \dots + 52000 + 9 \times 2500$$

$$(T_{18} - T_0)A_2 = 74500 + 1 \times 3500 + \dots + 74500 + 8 \times 3500$$

(using $S_n = \frac{n}{2}(a + L)$)

$$A_1 = \frac{10}{2}(52000 + 74500) = 632500$$

$$A_2 = \frac{9}{2}(74500 + 102500) = 722000$$

$$\text{Total} = 632500 + 722000 = \$1354500$$

b. (i) using $A = P(1 + \frac{r}{100})^n$ - Repayment

$$A_1 = 80000(1 + 0.02)^1 - M = 80000(1.02) - M \text{ as req.}$$

$$\begin{aligned} \text{(ii)} \quad A_2 &= A_1(1.02) - M \\ &= [80000(1.02) - M](1.02) - M \\ &= 80000(1.02)^2 - (1.02)M - M \\ &= 80000(1.02)^2 - M(1 + 1.02) \end{aligned}$$

$$\therefore A_3 = 80000(1.02)^3 - M(1 + 1.02 + 1.02^2) \quad \text{Total} = \$10385.57 + \$3649.27$$

$$\therefore A_{24} = 80000(1.02)^{24} - M(1 + 1.02 + \dots + 1.02^{23}) \text{ as required}$$

(iii) Now $A_{24} = 0$

$$\therefore M = \frac{80000(1.02)^{24}}{1 + 1.02 + \dots + 1.02^{23}}$$

Using $S_n = \frac{a(r^n - 1)}{r - 1}$ for series

$$S_{24} = \frac{1(1.02^{24} - 1)}{1.02 - 1} = 30.42186 \dots \text{ (calc)}$$

$$\therefore M = 4229.68778 \dots \text{ (calc)}$$

$\therefore$ Each repayment is \$4229.69

c. (i) Using $A = P(1 + \frac{r}{100})^n$

$$A_1 = 2000(1.04)^{42} = \$10385.57 \text{ as req.}$$

$$\text{(ii)} \quad A_2 = 250(1.04)^{40}$$

$$A_3 = 250(1.04)^{38}$$

$$A_4 = 250(1.04)^{36} \text{ and so on } \dots$$

$$\dots$$

$$A_{20} = 250(1.04)^2$$

$$A_{21} = 250$$

$\therefore A_1 + A_2 + A_3 + \dots + A_{21}$ forms a GP.

$$\begin{aligned} S_n &= \frac{a(r^n - 1)}{r - 1} \\ &= \frac{250 \times [(1.04)^{20} - 1]}{(1.04)^2 - 1} \\ &= \$3649.27 \end{aligned}$$

QUESTION 3.

a. $V = 240 \left(1 - \frac{t}{60}\right)^2$

(i) $60 = 240 \left(1 - \frac{t}{60}\right)^2$

$$\frac{1}{4} = \left(1 - \frac{t}{60}\right)^2$$

$$\pm \frac{1}{2} = 1 - \frac{t}{60}$$

$$\frac{t}{60} = 1 \pm \frac{1}{2}$$

$$\frac{t}{60} = \frac{3}{2}, \frac{1}{2}$$

$t = 90, 30$ outside time range

$\therefore$ bathtub is one quarter full after 30 seconds

(ii) $\frac{dv}{dt} = 480 \left(1 - \frac{t}{60}\right) x - \frac{1}{60}$
 $= -8 \left(1 - \frac{t}{60}\right)$

(When $t=30$)

$$\frac{dv}{dt} = -8 \left(1 - \frac{1}{2}\right) = -4$$

$\therefore$ the water is draining out at a rate of 4 litres/sec.

b. (i) Using Pythagoras' Theorem

$$x^2 = h^2 + 300^2$$

$$= h^2 + 90000$$

$$\therefore x = \sqrt{h^2 + 90000} \text{ as required}$$

(ii) to find $\frac{dx}{dt} = \frac{dx}{dh} \times \frac{dh}{dt}$

Now $x = (h^2 + 90000)^{1/2}$
 $\frac{dx}{dh} = \frac{1}{2} (h^2 + 90000)^{-1/2} \times 2h$
 $= \frac{h}{\sqrt{h^2 + 90000}}$

(When $h=400$)

$$\frac{dx}{dh} = \frac{400}{\sqrt{400^2 + 90000}}$$

$$= \frac{400}{500}$$

$$= \frac{4}{5}$$

$$\therefore \frac{dx}{dt} = \frac{4}{5} \times 5 = 4$$

the distance is increasing at 4m/s.

c. (i) If $N = 500 + Ae^{-kt}$ — ①

$$\frac{dN}{dt} = -kAe^{-kt}$$

$$= -k(N-500) \text{ as req.}$$

Since $N-500 = Ae^{-kt}$ from ①

(ii) when $t=0$ $N=3500$

$$\therefore 3500 = 500 + A$$

$$\therefore A = 3000$$

(When $t=3$) $3300 = 500 + 3000e^{-3k}$

$$2800 = 3000e^{-3k}$$

$$\frac{14}{15} = e^{-3k}$$

$$\ln\left(\frac{14}{15}\right) = -3k$$

$$-\frac{1}{3} \ln\left(\frac{14}{15}\right) = k$$

$$\therefore k = 0.022997623... \text{ (calc)}$$

$$\therefore k = 0.023 \text{ (3d.p.)}$$

(iii) $2300 = 500 + 3000e^{-kt}$

$$1800 = 3000e^{-kt}$$

$$\frac{3}{5} = e^{-kt}$$

$$\ln\left(\frac{3}{5}\right) = -kt$$

$$-\frac{1}{k} \ln\left(\frac{3}{5}\right) = t$$

$$\therefore t = 22.2121045... \text{ (calc)}$$

$\therefore$ after 22 years

QUESTION 4

a. (i) $v = \sqrt{4x^5 - 20x^3 + 40x}$

$$a = \frac{dv}{dx} \left(\frac{1}{2}v^2\right)$$

$$= \frac{d}{dx} \left(\frac{1}{2}(4x^5 - 20x^3 + 40x)\right)$$

$$= \frac{d}{dx} (2x^5 - 10x^3 + 20x)$$

$$= 10x^4 - 30x^2 + 20$$

(ii) Max. Speed occurs $a=0$

$$\text{i.e. } 10x^4 - 30x^2 + 20 = 0$$

$$x^4 - 3x^2 + 2 = 0$$

$$(x^2 - 2)(x^2 - 1) = 0$$

$$\therefore x = \pm\sqrt{2}, \pm 1$$

Max. speed occurs when $|v|$ is greatest.

(motion not possible if $x = -\sqrt{2}, -1$)

$$\therefore x = 1 \Rightarrow v = 2\sqrt{6}$$

$$x = \sqrt{2} \Rightarrow v = 4\sqrt{2}$$

Since $2\sqrt{6} > 4\sqrt{2}$, Max speed is $2\sqrt{6}$.

b. (i) $\frac{1}{2}v^2 = \frac{1}{2}x^2 - \frac{9}{2}$ ($a = \frac{d}{dx}(\frac{1}{2}v^2)$)

$$\therefore a = x$$

(ii) ($x=0$) $v^2 = 0 - 9$ $\therefore v^2 = -9$ which can't happen. $\therefore$ the particle does not pass through the

(iii) For motion to exist $v^2 \geq 0$

$$\therefore x^2 - 9 \geq 0$$

$$x \leq -3 \text{ or } x \geq 3$$

but when $t=0$, $x=5$ $\therefore x \leq -3$ is not possible and so $x \geq 3$ are the possible values.

(iv) The particle starts 5m to the right of and is moving towards 0. It continues to move to the left until it reaches $x=3$ where it stops ($v=0$). It then moves to the right and speeds up ($a>0$) continuing to move to the right.

c. (i) $a = -16e^{-4x}$

$$\frac{1}{2}v^2 = -16 \int e^{-4x} dx$$

$$\therefore \frac{1}{2}v^2 = 4e^{-4x} + C_1$$

$$(x=0, v=2\sqrt{2}) \quad 4 = 4 + C_1 \quad \therefore C_1 = 0$$

$$v^2 = 8e^{-4x} \quad \therefore v = \pm 2\sqrt{2}e^{-2x}$$

(Since $x=0, v=2\sqrt{2}$) $\therefore v = 2\sqrt{2}e^{-2x}$ as req.

(ii) $\frac{dx}{dt} = 2\sqrt{2}e^{-2x}$

$$\frac{dt}{dx} = \frac{e^{2x}}{2\sqrt{2}}$$

$$t = \frac{e^{2x}}{4\sqrt{2}} + C_2$$

($t=0, x=0$) $\therefore C_2 = -\frac{1}{4\sqrt{2}}$

$$\left\{ \begin{array}{l} t = \frac{e^{2x}}{4\sqrt{2}} - \frac{1}{4\sqrt{2}} \\ 4\sqrt{2}t = e^{2x} - 1 \\ 4\sqrt{2}t + 1 = e^{2x} \\ \ln(4\sqrt{2}t + 1) = 2x \\ \therefore x = \frac{1}{2} \ln(4\sqrt{2}t + 1) \text{ as req.} \end{array} \right.$$